Dataset with brain atlas A: use the Oasis dataset **for classification** of male and female individuals based on brain scans.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
yeo = datasets.fetch_atlas_schaefer_2018(n_rois=200)  # this needs to be modified for some tasks
print('%i regions found in this atlas' % (len(np.unique(nib.load(yeo.maps).get_data().ravel())) - 1))

masker = NiftiLabelsMasker(labels_img=yeo.maps, standardize=True,
    memory='nilearn_cache')
input_variables = masker.fit_transform(brain_data.gray_matter_maps)
output_variable = np.array(brain_data.ext_vars.mf == 'F', dtype=int)  # gives 1 for females and 0 for males
```

Task 1- Use LogisticRegression(penalty='l2', C=1.0) to fit separate classification models based on 100 brain regions of interest (n_rois=100). Perform dimensionality reduction on the training split via a) k-means clustering (n_clusters), b) GMMs (n_clusters), c) PCA (n_components), and d) ICA (n_components) to transform the original 100 input variables into k=1,2,3,…,10 latent pattern projections. Next, evaluate each fitted LogReg model on the unseen structural brain scans and retrieve classification accuracy. Do this prediction problem based on using 5-fold (KFold) cross-validation. Plot the classification accuracy on the test set vs k (k on the x-axis, test classification accuracy on the y axis; one line for each of the 4 dimensionality reduction methods).

Task 2- Same as 1 but use k=5 clusters/components and train a different LogisticRegression model on each atlas resolution (n_rois=100,200,…,1000). Plot the atlas resolution on the x-axis and test classification accuracy on the y-axis, and one line for each dimensionality reduction method.

Task 3- Combining knowledge from task 1 and 2, plot a heatmap that shows the atlas resolution (100-1000) on the x axis and the cluster/component number (10 choices) on the y axis, while the color of the tiles/cubes of the heatmap shows the final out-of-sample cross-validation estimate (classification accuracy). Since there are 4 different dimensionality reduction methods, there should be 4 such plots; one resolution-by-factor-number plot per clustering/decomposition method. Do another such 4 plots, but based on the average training set prediction accuracy (across the 5 training sets). You should have 8 plots overall, 2 for each reduction method.

Task 4- For each of the dimensionality reduction methods in Q1 (k-means, GMM, PCA, ICA) perform 5-fold cross-validation of the LogisticRegression model using k=10 clusters/components and the lowest atlas resolution (n_rois=100). In each fold use a

different random_state for the dimensionality reduction method and save the LogisticRegression model coefficients. Repeat this 5-fold cross-validation 20 times, each time using a different random_state for the KFold splitter to ensure the train/test split will be different.
For each of the four dimensionality reduction methods make a separate plot of the model coefficients, with the 10 coefficients on the x-axis and the corresponding coefficient value on the y-axis (there will be (20 KFold)*(5 fold) = 100 different estimates for each model coefficient). [think about this result obtained from these 4 resulting plots, and how this comes about given what you learned in the lecture]

Task 5- Same as task 4, but using a different random_state in each cross-validation fold and a fixed random_state for the dimensionality reduction methods. [think about why and how this is different from the 4 plots you got from task 5]

Dataset with brain atlas B: use the Oasis dataset to predict a **continuous outcome** (subject's age) from structural brain scans.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
from sklearn.preprocessing import StandardScaler
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
crad = datasets.fetch_atlas_craddock_2012()
atlas_nii = index_img(crad['scorr_mean'], 42) # this needs to be modified for some tasks
print('%i regions found in this atlas' % (len(np.unique(atlas_nii.get_data().ravel())) - 1))

masker = NiftiLabelsMasker(labels_img= atlas_nii, standardize=True)
input_variables = masker.fit_transform(brain_data.gray_matter_maps)
output_variable = StandardScaler().fit_transform(brain_data.ext_vars.age[:, None])[:, 0] #
gives subject age on standard units after z-scoring
```

Note: use Ridge() instead of LogisticRegression() to predict the continuous subject age using the lowest-resolution version of Craddock region atlas (.scorr_mean), and use the R2 coefficient of determination instead of classification accuracy (*sklearn.metrics.r2_score).*

6- analogous to task 1

7- analogous to task 2

8- analogous to task 3

9- analogous to task 4

10- analogous to task 5